

Thomas Collins

Spatial Data Analyst & Postdoctoral Researcher

+44-7563-982-877

trc207 [at] exeter.ac.uk

📍 Department of Computer Science, University of Exeter (UoE), UK



Professional Summary

Analytical researcher and spatial data specialist with experience designing and delivering complex intelligence and risk-analysis projects using behavioural, environmental, demographic, and geospatial datasets. Skilled in statistical modelling, GIS, machine learning, and evidence-led analysis to support strategic planning and decision-making.

Experienced in translating complex analytical findings into clear, actionable outputs for technical, policy, and non-technical audiences, with a strong understanding of data governance, ethical data use, and interdisciplinary collaboration. Motivated to apply advanced analytical and strategic intelligence skills within a public sector environment supporting community resilience and risk-informed decision-making across Cornwall.

Technical Skills

- **Data Analysis & Modelling:** Python, R, statistical modelling, machine learning, spatial statistics, geospatial analysis, multivariate analysis
- **Geospatial & Remote Sensing:** OpenStreetMap, Google Earth Engine, remote sensing workflows, spatial data integration, GIS
- **Data Visualisation & Reporting:** Jupyter, L^AT_EX, technical reporting, analytical visualisation, scientific documentation
- **Development & Workflow Tools:** GitHub, Conda, PyCharm, Overleaf, Visual Studio
- **Cloud & Command Line Tools:** Amazon Web Services (AWS), Bash, PowerShell, Windows Command Prompt

Current Position

Postdoctoral Researcher

Centre for Environmental Intelligence, UoE, Exeter

OCT 2025 -- CURRENT

- Designing and leading complex spatial and statistical analyses integrating behavioural, environmental, demographic, and remotely sensed datasets to support strategic environmental planning and evidence-led decision making.
- Developing analytical frameworks to assess environmental risk, model the impacts of development scenarios on protected sites and communities, and identify future pressures relevant to long-term planning and policy.
- Producing technical reports, visualisations, methodology documents, and analytical briefings communicating complex findings clearly to government agencies, local authorities, and interdisciplinary partners.
- Providing specialist advice on analytical methodologies, data governance, and the ethical integration of spatial, behavioural, and environmental intelligence within policy-focused projects.
- Collaborating with Natural England, local planning authorities, and interdisciplinary research teams to deliver integrated analytical outputs aligned with national environmental policy frameworks.
- Managing multiple analytical workstreams simultaneously, ensuring outputs are delivered to agreed timescales, quality standards, and stakeholder requirements.

Previous Professional Experience

• **PhD Researcher, University of Exeter**

Oct 2021 - Oct 2025

Exeter, UK

- Designed and delivered advanced analytical workflows integrating large-scale mobile phone mobility data, app usage data, remote sensing, geospatial systems, and machine learning techniques to model urban behaviour, movement patterns, and spatial dynamics.
- Developed spatial and statistical modelling frameworks, including Elastic Net, Random Forests, and geographically weighted modelling approaches, to identify environmental, infrastructural, and behavioural drivers of inequality, vulnerability, and urban activity.
- Produced analytical reports, technical outputs, visualisations, publications, and conference presentations communicating complex findings to technical, policy, and non-technical audiences.
- Managed large and complex datasets while maintaining reproducible analytical workflows, transparent documentation, and robust quality assurance processes.

- Collaborated across interdisciplinary teams spanning computer science, geography, planning, and statistics to deliver integrated analytical outputs aligned with research and policy objectives.
 - Developed novel analytical approaches to improve the robustness, interpretability, and strategic relevance of spatial and behavioural intelligence outputs.
 - Managed multiple concurrent analytical workstreams, prioritising tasks and delivering outputs within demanding project deadlines.
- **Research Data Officer, Royal Cornwall Hospitals Trust** *Oct 2021 - Feb 2023*
Treliske, Truro
- Supervised and supported the Assistant Research Data Officer and delegated tasks effectively to support haematological cancer research.
 - Ensured accurate and timely data management across studies, coordinating with study leads, the haematology clinical team, and external stakeholders.
 - Managed sensitive clinical research datasets and resolved complex data queries to maintain data integrity, regulatory compliance, and operational accuracy.
 - Supported the Senior Research Data Officer in aligning data management practices with regulatory guidelines, Trust Policies, and standard operating procedures.
 - Applied robust data validation, quality assurance, and governance processes in line with Good Clinical Practice standards.
 - Supported research governance processes including ethics submissions, sponsor communications, and compliance with HRA and Trust procedures.
 - Completed mandatory training in Good Clinical Practice (GCP) and data security as part of NHS RD governance compliance.

Education

PhD in Computer Science	Department of Computer Science, UoE, Exeter	OCT 2021 – OCT 2025
--------------------------------	---	---------------------

- Research focused on urban environments, spatial inequality, environmental exposure, and urban vibrancy using computational social science and geospatial analysis approaches.
- Applied machine learning, GIS, remote sensing, and behavioural data analysis techniques to investigate interactions between people and urban environments.
- Produced peer-reviewed publications and conference presentations communicating research findings across interdisciplinary academic and policy contexts.

M.Sci. with honours in Conservation Biology and Ecology	Centre for Ecology and Conservation, UoE, Exeter	SEP 2017 – MAY 2021
---	--	---------------------

- Specialised in spatial ecology, movement ecology, behavioural analysis, and environmental data analysis.
- Conducted independent research projects involving GPS tracking, spatial modelling, and behavioural analysis within urban and ecological systems.

Projects

-
- **‘Pathways to Digital Innovation’ training and internship** *Jun 2018*
UoE, Penryn Campus
- Earned a place on the Pathways to Digital Innovation Programme, a Professional Pathways initiative focusing on careers involving Big Data, Geographic Information Systems (GIS), and technology.
 - Participated in a comprehensive three-day training session led by industry experts as part of the program and gained valuable sector-specific insights through engaging talks, workshops, and projects during the training session.

Certifications & Professional Training

-
- **Full Accredited Researcher under the Digital Economy Act 2017 (DEA)**
Office for National Statistics
 - **Good Clinical Practice (GCP)**
National Institute for Health and Care Research (NIHR)
 - **Information Governance and Data Security Awareness**
NHS Research & Development Governance Training
 - **Learning and Teaching in Higher Education (LTHE)**
AdvanceHE

Recommendations

Recommendations are available on request but also see [HERE](#).